

Stochastic gradient descent optimisers

A cheat sheet of stochastic gradient descent optimisers

Last updated: 22 Sep 2019

Contents

(sorted by publication date)

- [Vanilla SGD](#)
- [Momentum](#)
- Averaged SGD
- [AdaGrad](#)
- [Adadelta](#)
- [RMSProp](#)
- [Nesterov-accelerated gradient](#)
- [Adam](#)
- [AdaMax](#)
- [Nadam](#)
- [SGDW](#)
- [AdamW](#)
- LARS
- [SWA](#)
- [AMSGrad](#)
- LAMB
- [NovoGrad](#)
- [RAdam](#)

CATEGORIES

Stochastic gradient descent (SGD) optimisers can be broadly categorised into (1) **adaptive** learning rate schemes (e.g. Adam and RMSProp) and (2) **accelerated** schemes (e.g. Nesterov) ^[1].

NOTATION

- w_t — weight at time step t
- g_t — gradient w.r.t. w at time step t

- α — learning rate
- λ — weight decay parameter
- ϵ — a very small number, used to prevent division by zero
- m — estimate for first moment of gradient
- v — estimate for second moment of gradient
- $^{(l)}$ — specific to layer l
- $|| \cdot ||$ — absolute value
- $|| \cdot ||_2$ — Euclidean/L2 norm

Vanilla SGD

$$w_{t+1} = w_t - \alpha g_t$$

Momentum [2]

$$w_{t+1} = w_t - \alpha m_t$$

where

$$m_t = \beta m_{t-1} + (1 - \beta) g_t$$

AdaGrad (adaptive gradient) [3]

$$w_{t+1} = w_t - \frac{\alpha}{\sqrt{v_t + \epsilon}} \cdot g_t$$

where

$$v_t = v_{t-1} + [g_t]^2$$

Adadelta [4]

$$w_{t+1} = w_t - \frac{\sqrt{D_{t-1} + \epsilon}}{\sqrt{v_t + \epsilon}} \cdot g_t$$

where

$$D_t = \beta D_{t-1} + (1 - \beta) [\Delta w_t]^2$$
$$v_t = \beta v_{t-1} + (1 - \beta) [g_t]^2$$

RMSProp [5]

$$w_{t+1} = w_t - \frac{\alpha}{\sqrt{v_t + \epsilon}} \cdot g_t$$

where

$$v_t = \beta v_{t-1} + (1 - \beta) [g_t]^2$$

Nesterov-accelerated gradient [6]

$$w_{t+1} = w_t - \alpha m_t$$

where

$$m_t = \beta m_{t-1} + (1 - \beta) g_*$$

Note that g_* is the gradient of w_* , where

$$w_* = w_t - \alpha m_{t-1}$$

Adam (adaptive moment estimation) [7]

$$w_{t+1} = w_t - \frac{\alpha}{\sqrt{\hat{v}_t} + \epsilon} \cdot \hat{m}_t$$

where

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$$
$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

and

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$
$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) [g_t]^2$$

AdaMax [7]

$$w_{t+1} = w_t - \alpha \cdot \frac{\hat{m}_t}{s_t}$$

where

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$$

and

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$
$$s_t = \max(\beta_2 s_{t-1}, |g_t|)$$

Nadam (Nesterov-accelerated Adam) [8]

$$w_{t+1} = w_t - \frac{\alpha}{\sqrt{\hat{v}_t} + \epsilon} \cdot \beta_1 \hat{m}_t - \frac{1 - \beta_1}{1 - \beta_1^t} \cdot g_t$$

where

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$$
$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

and

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$
$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) [g_t]^2$$

SGDW (SGD with decoupled weight decay) [9]

Note: SGDW is equivalent to optimising a loss function with L2 regularisation using SGD with momentum *and* weight decay (see [appendix](#)). Scheduler rate is set to 1 for readability.

$$w_{t+1} = w_t - \alpha m_t - \lambda w_t$$

where

$$m_t = \beta m_{t-1} + (1 - \beta) g_t$$

AdamW (Adam with decoupled weight decay) [9]

Note: Here the scheduler rate is set to 1 for readability.

$$w_{t+1} = w_t - \frac{\alpha}{\sqrt{\hat{v}_t} + \epsilon} \cdot \hat{m}_t - \lambda w_t$$

where

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

and

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) [g_t]^2$$

SWA (stochastic weight averaging) [10]

Apart from

$$w_{t+1} = w_t - \alpha g_t$$

the optimiser maintains another set of weights m which will be used in the final model. The update of m is:

$$m_{t+1} = \begin{cases} \beta_t m_t + (1 - \beta_t) w_t & \text{if } \text{mod}(t, c) = 0 \\ m_t & \text{otherwise} \end{cases}$$

where m is updated with the running average every c time steps and

$$\beta_t = \frac{t}{t + c}$$

The equations look different from the paper but are essentially identical due to my reparametrisation (see [appendix](#)).

AMSGrad^[11]

$$w_{t+1} = w_t - \frac{\alpha}{\sqrt{\hat{v}_t} + \epsilon} \cdot m_t$$

where

$$\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$$

and

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) [g_t]^2 \end{aligned}$$

NovoGrad^[12]

Note that weight update is done per layer, indicated by the superscript $^{(l)}$.

$$w_{t+1}^{(l)} = w_t^{(l)} - \alpha m_t^{(l)}$$

where

$$m_t^{(l)} = \beta_1 m_{t-1}^{(l)} + (1 - \beta_1) \left(\frac{1}{\sqrt{v_t^{(l)}} + \epsilon} \cdot g_t^{(l)} + \lambda w_t^{(l)} \right)$$

and

$$v_t^{(l)} = \beta_2 v_{t-1}^{(l)} + (1 - \beta_2) \|g_t^{(l)}\|_2$$

RAdam (Rectified Adam) [13]

$$w_{t+1} = w_t - \frac{\alpha}{\sqrt{\hat{v}_t} + \epsilon} \cdot r_t \hat{m}_t$$

where

$$\begin{aligned}\hat{m}_t &= \frac{m_t}{1 - \beta_1^t} \\ \hat{v}_t &= \begin{cases} \frac{v_t}{1 - \beta_2^t} & \text{if } p_t > 4 \\ 1 & \text{otherwise} \end{cases} \\ r_t &= \begin{cases} \sqrt{\frac{(p_t-4)(p_t-2)p_\infty}{(p_\infty-4)(p_\infty-2)p_t}} & \text{if } p_t > 4 \\ 1 & \text{otherwise} \end{cases}\end{aligned}$$

and

$$\begin{aligned}m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) [g_t]^2 \\ p_t &= p_\infty - \frac{\beta_2^t}{1 - \beta_2^t} \cdot 2t \\ p_\infty &= \frac{2}{1 - \beta_2} - 1\end{aligned}$$

Updates and corrections

If you see mistakes or want to suggest changes, kindly email me at raimi.bkarim@gmail.com

Weight decay [14]

$$w_{t+1} = (1 - \lambda)w_t - \alpha g_t$$

Weight decay and L2

SGD with L2 regularisation is the same as SGD with weight decay, as proven in [9].

For a loss function with L2 regularisation and regularisation parameter λ' , we have:

$$f(w) = L(w) + \frac{1}{2}\lambda' \|w\|_2^2$$

and its corresponding gradient descent weight update is:

$$\begin{aligned}w_{t+1} &= w_t - \alpha \frac{\partial f}{\partial w} \\&= w_t - \alpha \frac{\partial L}{\partial w} - \alpha \lambda' w_t\end{aligned}$$

For SGD with weight decay λ but no regularisation, the gradient update is

$$\begin{aligned}w_{t+1} &= (1 - \lambda)w_t - \alpha \frac{\partial L}{\partial w} \\&= w_t - \alpha \frac{\partial L}{\partial w} - \lambda w_t\end{aligned}$$

Here we can see that λ is just a reparametrisation of $\lambda = \alpha \lambda'$.

SWA reparametrisation

From the pseudo code in the paper and taking $n_{\text{models}} = n$, the update is

$$\begin{aligned}w_{\text{SWA}} &= \frac{w_{\text{SWA}} \cdot n + w}{n + 1} \\&= \frac{n}{n + 1}w_{\text{SWA}} + \frac{1}{n + 1}w \\&= \frac{n}{n + 1}w_{\text{SWA}} + \frac{(n + 1) - n}{n + 1}w \\&= \frac{n}{n + 1}w_{\text{SWA}} + \left(1 - \frac{n}{n + 1}\right)w \\&= \beta w_{\text{SWA}} + (1 - \beta)w\end{aligned}$$

where $\beta = \frac{n}{n+1}$. And because $n = n_{\text{models}} = \frac{t}{c}$,

$$\beta = \frac{t/c}{t/c + 1} = \frac{t/c}{(t + c)/c} = \frac{t}{t + c}$$

Related

<https://towardsdatascience.com/10-gradient-descent-optimisation-algorithms-86989510b5e9>

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<https://keras.io/optimizers/>

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